



Classification of patients by severity grades during triage in the emergency department using data mining methods

Dror Zmiri MSc,¹ Yuval Shahar MD PhD² and Meirav Taieb-Maimon PhD³

¹Graduate Student, Medical Informatics Research Center, ²Professor and Head of the Medical Informatics Research Center, ³Lecturer, Department of Information Systems Engineering, Ben Gurion University, Beer Sheva, Israel

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Correspondence

Mr Dror Zmiri
42/10 Nachal Dolev Street
Beit Shemesh 99621
Israel
E-mail: zmiridro@bgu.ac.il

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Abstract

Objective To test the feasibility of classifying emergency department patients into severity grades using data mining methods.

Design Emergency department records of 402 patients were classified into five severity grades by two expert physicians. The Naïve Bayes and C4.5 algorithms were applied to produce classifiers from patient data into severity grades. The classifiers' results over several subsets of the data were compared with the physicians' assessments, with a random classifier, and with a classifier that selects the maximal-prevalence class.

Measurements Positive predictive value, multiple-class extensions of sensitivity and specificity combinations, and entropy change.

Results The mean accuracy of the data mining classifiers was $52.94 \pm 5.89\%$, significantly better ($P < 0.05$) than the mean accuracy of a random classifier ($34.60 \pm 2.40\%$). The entropy of the input data sets was reduced through classification by a mean of 10.1%. Allowing for classification deviations of one severity grade led to mean accuracy of $85.42 \pm 1.42\%$. The classifiers' accuracy in that case was similar to the physicians' consensus rate. Learning from consensus records led to better performance. Reducing the number of severity grades improved results in certain cases. The performance of the Naïve Bayes and C4.5 algorithms was similar; in unbalanced data sets, Naïve Bayes performed better.

Conclusions It is possible to produce a computerized classification model for the severity grade of triage patients, using data mining methods. Learning from patient records regarding which there is a consensus of several physicians is preferable to learning from each physician's patients. Either Naïve Bayes or C4.5 can be used; Naïve Bayes is preferable for unbalanced data sets. An ambiguity in the intermediate severity grades seems to hamper both the physicians' agreement and the classifiers' accuracy.

Introduction

Triage is the process of initial classification of patients in accordance with the type of injury they have incurred (internal, orthopaedic, surgical, etc.) and the severity grade of that injury, based on medical information (vital signs such as blood pressure, patient complaints such as chest pain, and demographic data such as age) available at that initial stage. Successful triage enables the prioritization of the patients according to medical criteria to render the most efficient use of the medical resources available.

The crucial need for initial classification of patients into severity grades as part of the triage process motivates the search for a general method to create algorithms capable of doing so, in a manner similar to that of expert physicians who do it. Such a method can support automated patient classification into severity grades.

In this paper, we will present a method based on a data mining (DM) approach, in which the classification model is learned from examples of patient classification performed by physicians.

Background

The growth in the number of people turning to EDs for emergency medical care is a continuous and global phenomenon [1,2].

Methods for performing triage

Most of the existing EDs in the world use the assistance of registered nurses to direct patients to the appropriate section of the ED for treatment, possibly classifying them according to the severity of their conditions by following certain guidelines or flow charts [3,4].

Other medical centres reported using non-professional personnel at the triage station, aided by a computerized system that is based on rules developed manually. These centres often expressed satisfaction with the results [1,5,6].

The methods of triage in use today rank patient cases by their severity on a three-, four-, or five-grade scale. Most American EDs use a three-grade scale, while Canadian, British and Australian EDs all shifted to a five-grade scale [7–9].

Several methods have been developed for the prioritization of incoming patients by degree of severity. Some classify patients by their complaint or their main problem, while others follow rules developed manually [10,11]. The Emergency Severity Index system classifies patients by their conditions, vital signs, and an assessment of the resources necessary to diagnose and treat the patient [12]. Another method, developed by the Manchester Triage Group, uses different algorithms for each major complaint, such that, except for the general questions common to all complaints, each algorithm is characterized by special questions, suited to specific complaints [13].

In a study in which nurses were asked to rank the severity of 50 different cases (presented to them in writing) using the Canadian method, a high degree of agreement was found amongst the nurses [9]. In another study, using a triage scale with five grades of severity and an appropriate training method proved to be very reliable [10]. Good results, with respect to inter-observer agreement, were also attained when evaluating other methods with five grades of severity, such as the Canadian or Emergency Severity Index methods [12].

Other studies, however, found a low to medium level of agreement among the medical personnel. In a study dealing with the classification of patients by nurses compared with that performed by physicians using four severity grades, only a medium level of agreement was found [14]. Similar results were attained in a study that tested the agreement between various medical personnel, when asked to assign patients' records to either an 'urgent' or a 'non-urgent' category [15], and in a study based on a British scale with four severity grades [16]. One study, based on 1213 incoming ED patients, evaluated a four-grade triage scale, and found only a disappointing 18% agreement rate between the nurses and the physicians [17]. A study using a three-grade severity triage scale also showed low reliability [18].

To summarize, studies show different degrees of validity of the classification by severity grades. This may be because of the difficulty of developing a flow chart which can sufficiently take into account all of the potential combinations of the different symptoms. It is likely that one of the causes influencing the level of validity of the scale is the amount of training in using the scale that the personnel received.

Triage decision support systems

Initially, triage had been based mostly on written instructions and algorithms. However, over the years, the use of computerized systems became more and more prevalent. In 1981, a study was published that described a computerized triage system run by unskilled personnel, yet found to be safe and reliable [6]. Later on, an efficient triage system was described, used in American military hospitals by paramedics, based on an algorithm applied to the main complaint [19]. In another study, a comparison was made

between triage performed by nurses and by a computer, and it was found that the results of the computer's triage were more accurate [5]. In another study, however, computerized triage was found to be inaccurate, when compared with that of a physician. Brillman *et al.* thought that computerized triage was not reliable enough, because of high sensitivity but low specificity [1].

As far as we know, no study similar to ours has been done thus far. Namely, a study in which the classification of patients into severity grades was performed by means of DM, based on the severity grades determined by expert physicians for a set of patients. Most methods for triage are based on algorithms created manually by a consensus of a group of physicians or medical personnel, who determine the severity grade for each hypothetical case, such as done in the Canadian method [7], or in the method developed by the Manchester Triage Group [13]. Obviously, developing such systems takes much time and many resources, in particular a significant time investment by physicians, who must both define the algorithm and maintain it. However, when using a computer system that learns automatically, most of the human effort is centred on classifying a sufficient number of actual cases for training purposes, but otherwise there is very little investment in creating the knowledge base, and its maintenance is essentially automatic, the knowledge continuously improving with the entry of new classified cases.

The goal of the current study was to develop a method for designing a classifier (i.e. an algorithm generated by DM methods) that can automatically determine the severity grade of patients for purposes of triage in hospital EDs given the patients' characteristics. The objective of this method is to learn from classified patient cases, and using the severity grades provided by the physicians, to generate a classifier able to classify new patient into respective severity grades, just as the physicians would have done, given the same information. Thus the gold standard for testing these classifiers is the decision the physicians would make for the same patients.

Research questions

Given the goals described above, our research questions were therefore:

- 1 Can the classifiers we generate using DM methods perform significantly better than a classifier using just the distribution of the severity grades, within the cases classified by the expert physicians?
- 2 Can our method reduce the level of uncertainty regarding the distribution of severity grades within a given set of cases?
- 3 Does the number of severity grades influence the accuracy of the classifier?
- 4 Can a generated classifier predict the severity grade assigned to a patient by a physician, at least as well as another physician could predict that assignment (as indicated by their degree of agreement)?
- 5 Does learning from a set of patient records about which there was a consensus amongst physicians improve the generated classifier's accuracy, compared with learning from a set of records classified by only one physician?
- 6 Are there differences in performance amongst the various methods we used to generate classifiers?

Methods

The data

This research was carried out at the Soroka Academic Medical Center in Beer-Sheva, Israel. The ED at the Soroka Medical Center receives approximately 130 000 cases per year, of which about 20% of the patients are hospitalized for continued treatment in the many hospital departments.

Because of the different natures of the various medical cases, patients are referred to different parts of the ED. This paper deals only with those patients referred to the internal medicine section of the ED (in which they receive primary emergency care for internal diseases). This is the most overloaded section of the ED and receives around 45% of all the incoming ED patients. We also excluded those patients who were immediately shunted to the Trauma Unit, because they are generally taken directly to Trauma so quickly that the nurses do not do any formal triage processing at all.

The survey data were collected over four standard mid-weekdays (during the summer), throughout all the hours of the day. Data were collected in the ED that accepts only adult (older than 18 years.) patients. Overall, data were gathered regarding 402 incoming patients. The two most senior physicians in the ED participated in this study, and were asked to classify the original ED patient records (using only the data known on arrival to the ED) according to their severity grades. These decisions, regarding the degrees of severity, were being made retrospectively by both physicians, based on the written data from the cases, as recorded at the time the patients were admitted into the ED. These data included: age, sex, vital signs (e.g. temperature, pulse, blood pressure, respiration, saturation and glucose), the main complaint made by the patient, previous diagnoses known at the time of arrival to the ED, and a letter of referral to the ED.

One physician ('EX1') classified 402 cases while the other physician ('EX2') classified only 187 of them. (Thus, the first physician classified an extra 215 cases.) This classification process served as the 'gold standard' for all the cases in which an evaluation was made of the results provided by the algorithm, and we refer to this when we discuss the 'actual severity grade'. It should be noted that recruiting two senior physicians for the task of manual classification of hundreds of ED cases was not trivial, and the number of classified records is thus relatively high, in terms of studies involving a human-based gold standard.

Methods of analysis

As explained above (in section 'Methods for performing triage'), no international standard exists for the classification of patients into severity grades. Moreover, at the medical centre in which we carried out our study (as in many other medical centres), there is no formal triage process for classifying the incoming patients into severity grades, although there is willingness to adopt a five-grade severity scale and there exists an informal agreement regarding the semantics determining those grades. Therefore, for our research, we decided to use a technique taken from DM, which attempts to learn the hidden decision-making algorithm used by experienced physicians, when applied to the real case-data and in accordance with their predetermined classification scale.

Taking into account that, although the physicians were asked to rank the severity grade of each patient on a scale of 1–5, in practice, none of the 402 incoming patients were given a grade-5 ranking. (Grade 5 is meant to be given to patients in critical condition, taken directly into the Trauma Unit, who actually skip the triage process.) Thus, we amended our decision and adopted a more practical severity scale of 1–4.

To learn how the physicians ranked the severity grades of the ED patients' cases during triage, and to define our 'gold standard', we employed two different types of DM. We chose both Naïve Bayes model and C4.5 DM algorithms to help us create our classifiers for severity grades. Both these algorithms represent independent methodologies for creating classifiers, and represent different families of algorithms: decision trees and probabilistic algorithms. To run the Naïve Bayes model and the C4.5 algorithms we used the Weka library, 3.4.5 version, created at the University of Waikato in New Zealand [20]. To implement C4.5 we used J48, the Weka version of the algorithm C4.5.

Evaluation methods

To assess the validity of the DM process, we used a 10-fold cross-validation.

To compare our results with those achieved by non-learning classification models, which use only the initial class (i.e. severity level in this case) distribution, we assessed them versus the following classifiers:

Random Classifier – a classification model that provides a severity grade for all the cases that have been classified, in proportion to the known distribution of severity grades. (That is, a severity grade is assigned to a case with a probability equal to its proportion in the study population, as determined by the physician-based classification). This method is possible when all that is known is the distribution of the severity grades in the population, and the goal is to avoid missing one of the severity grades entirely.

Maximum probability classifier – a classification model that assigns to all cases the most prevalent severity grade in the study population, as determined by the physician(s). Thus, all cases will be assigned the same severity grade.

To compare the percentage of hits attained under different circumstances (various algorithms, different severity grades, Consensus regarding all the cases, etc.), we applied statistical proportions tests.

Evaluation measures

As a standard by which to measure the significance and validity of our results, we used multiple-value (positive) predictive accuracy, that is, the percentage of cases in which the ranking by severity (from 1 to 4) matched the severity grade as determined for the same cases by the physicians (later we use the term 'accuracy' to represent this standard).

Multiple value (positive) predictive Accuracy =

$$\sum_{i=1}^4 P(i|i')P(i')$$

[$P(i')$ represents the probability that the algorithm will yield a classification of class i for a given case, and $P(i)$ represents the probability that the actual severity grade is i].

However, as the distribution of the severity grades is unbalanced, and as in many cases, a high accuracy rate is insufficient, and the ability to detect severity grades that have a low prevalence is crucial, especially in a medical domain, we applied additional measures, such as those that incorporate both sensitivity and specificity [21]. These additional measures can highlight, for example, the fact that a classifier might achieve an overall relatively high accuracy, while actually completely ignoring one or more classes.

Note that sensitivity and specificity are dependent on the severity grade. The *sensitivity* expresses the probability that the classifier will predict the class i [we indicate this as $P('i'|i)$], when the actual grade is, in fact, i :

$$\text{Sensitivity}(i) = P('i'|i)$$

Note that this definition is the same, whether only two classes exist, or more.

Specificity expresses the probability that the classifier will predict that the grade is not i , when the actual grade is, in fact, not i [i.e. $P('i'|i)$]. Thus in the common binary situation of two classes C_1 & C_2 , $\text{specificity}(1) = P('2'|2)$. In our study, as the number of severity grades is greater than two, we have extended the notion of binary specificity to multiple-value specificity (specificity*) as follows:

$$\text{Specificity}^*(i) = \frac{\sum_{j \neq i} \sum_{k \neq i} P('j'|k)P(k)}{1 - P(i)} = \frac{\sum_{j \neq i} \sum_{k \neq i} P('j' \cap k)}{\sum_{j \neq i} P(j)}$$

Note that this definition converges to the usual binary definition of specificity when there are only two classes. Thus, we will refer to this definition simply as 'specificity'.

Because of the fact that there is a trade-off between the sensitivity and the specificity measures, and there is great significance, especially in medicine, to the identification of the rarer cases, and not just to achieve higher accuracy (which can often be achieved by using a naïve classification model, such as Maximum probability classifier) – we must use other measures, which integrate both of these basic measures.

The first such weighted measure, applied to sensitivity and specificity, is the Balanced Correctness Rate (BCR), defined as:

$$\text{BCR} = \frac{\text{Sensitivity} + \text{Specificity}}{2}$$

The BCR measure is equivalent to the Balanced Error Rate measure, often used in previous studies (e.g. by Cho *et al.* [21]), because they complement one another to the value of 1. In light of the fact that patients were classified to severity grades, and that each severity grade has its own sensitivity and specificity, we used the multiple-value BCR (BCR*) measure, which serves as an enhancement of the BCR measure for cases in which the target variable (in our case, the severity grade) has more than two possible values. BCR* is defined as follows:

$$\text{BCR}^* = \frac{\sum_{i=1}^N \text{BCR}_i}{N}$$

where BCR_i uses the extension of Sensitivity(i) and Specificity(i) as above.

Another measure that we used, which was also used in previous studies (e.g. by Cho *et al.* [21]), is the Harmonic Mean of Sensitivity and Specificity (HMSS):

$$\text{HMSS} = \frac{2 * \text{Sensitivity} * \text{Specificity}}{\text{Sensitivity} + \text{Specificity}}$$

HMSS is more sensitive to small changes in the sensitivity or in the specificity, especially when one of them equal to zero.

We used the multiple-value HMSS (HMSS*) measure, which serves as an enhancement of the HMSS measure for cases in which the target variable (in our case, the severity grade) has more than two possible values.

$$\text{HMSS}^* = \frac{\sum_{i=1}^N \text{HMSS}_i}{N}$$

Where HMSS_i uses the extension of Sensitivity(i) and Specificity(i) as above.

A useful measure we have used to assess multiple class separation was the amount by which the classifier reduced the uncertainty regarding the appropriate severity grade by using the Entropy measure:

$$\text{Entropy}(S) = \sum_{i=1}^k -P_i \log_2 P_i$$

Where S is the number of cases (patient records), and P_i is the frequency of severity grade i for k possible grades.

Thus, we first calculate the prior Entropy (H_{prior}) by calculating the entropy of the original distribution in the population. Then we compute the posterior entropy ($H_{\text{posterior}}$) as follows: first, compute the entropy for each value ' i ' predicted by the classifier, by noting the distribution of the actual severity grades ($i = 1, \dots, k$) given prediction ' i '. Then summing the entropy for all the prediction ' i ' ($i = 1, \dots, k$), weighting the entropy for prediction ' i ' by the frequency of prediction ' i ' being made by the classifier.

In order to test whether the classifiers produced by the DM algorithms actually learn significantly, thus reducing the degree of uncertainty in the distribution of the severity grades of patients, we applied the $H_{\text{classifier}}$ measure which calculates the difference between the prior Entropy and the posterior Entropy:

$$H_{\text{classifier}} = H_{\text{prior}} - H_{\text{posterior}}$$

We also used the Relative Classifier Information (RCI) measure in the same fashion it was suggested by Sindhwani *et al.* [22]. The RCI measure calculates the percentage of change in the Entropy:

$$\text{RCI} = \frac{H_{\text{classifier}}}{H_{\text{prior}}}$$

Results

Description of the patient data

Data were gathered regarding 402 adult patients, of whom 204 were men (mean age: 47.03 ± 21.41) and 198 were women (mean age: 51.72 ± 21.93).

Categorization of the patient data

The classification of the 402 patient cases (or a subset of this group) into severity grades was performed, as stated above, by the two physicians EX1 and EX2, whose classifications were designated as the *gold standard*. The outcome of this process included several *labelled record sets*, that is, sets of patient records, each case labelled by a severity grade by one or both physicians.

We considered five labelled record sets, to be denoted as follows:

- 1 '402, EX1' – all 402 patient cases labelled by EX1.
- 2 '402, EX1 + EX2' – all 402 patients' cases labelled by both physicians, when the first 215 cases were labelled by EX1, and the other 187 cases were labelled by EX2.
- 3 '215, EX1' – the 215 patient cases labelled by EX1.
- 4 '187, EX1' – the 187 patient cases labelled by EX1.
- 5 '187, EX2' – the same 187 patient cases labelled by EX2.

(In the section 'Comparison of the results of the DM versus the agreement between the physicians', we take advantage of the fact that 187 patients' cases were labelled by both physicians.)

Analysis of the DM results

The accuracy of the classifiers generated by the two DM algorithms was compared with the accuracy of the Random Classifier, using a proportions test (Table 1).

As shown in Table 1, there is a significant difference between the accuracy rates produced by the DM-generated classifiers and those of the Random Classifier, with higher accuracy for the DM-generated classifiers.

(Note that the 95% confidence interval refers to the difference in the proportions; if it includes the 0 point the difference is insignificant).

A similar comparison (Table 2) was performed between the accuracy of the DM-generated classifier and those of the Maximum probability classifier. In this case, as the predominant severity grade assigned by the physicians to the greatest number of cases was grade 1, in all labelled record sets all patients were labelled as grade 1.

As can be seen in Table 2, despite the higher accuracy achieved in all the cases by the DM-generated classifiers, as opposed to the Maximum probability classifier, there is a significant difference only in some of the cases.

To complement the accuracy measure we performed a comparison among the DM-generated classifiers and the Random and Maximum probability classifier, using the BCR* and HMSS* measures (Table 3).

As can be seen in Table 3, the value of the BCR* measure in regard to the results of the Random Classifier and Maximum probability classifier was 0.5 for all the classified case categories. Yet the BCR* values obtained for the results of our DM-generated classifiers were higher than 0.5 for all the cases.

In regard to the HMSS* measure, the values derived from this measure for the DM-generated classifiers were higher than those of both the Random Classifier and the Maximum probability classifier in all the classified case categories. The value of the HMSS* measure obtained for the Maximum probability classifier is 0 for all the cases, because of a 0 sensitivity or a 0 specificity (depending on the particular severity grade), indicating the inferiority of this classification model as against the results produced by our DM-generated classifiers.

Reduction of the classification uncertainty

We tested whether the DM-generated classifiers *learn*, in the sense that following training on some of the cases, they were able to reduce the prior uncertainty about the rest of the cases regarding the severity grade. To perform this comparison, we used the $H_{\text{classifier}}$ and RCI measures (Table 4).

As can be seen in Table 4, the values of the $H_{\text{classifier}}$ and the RCI measure were greater than 0 for all the classified case categories and for the two DM-generated classifiers, that is, in all the cases, the calculated posterior Entropy was smaller than the prior Entropy. This means that the use of the algorithm reduced the level of uncertainty of the severity grade, in comparison with the original level of uncertainty (before its application). Note that both the Random Classifier and the Maximum

Table 1 A comparison of the accuracy of the classifiers generated by the data mining algorithms with the accuracy of the Random Classifier, using a proportions test

		Data mining-generated classifier		Random Classifier		z statistic	P-value	Confidence interval (95%)
		Accuracy (%)	n	Accuracy (%)	n			
Naïve Bayes [†]	402, EX1 + EX2	56.72	228	31.84	128	7.1*	0.000	[-0.315, -0.182]
	402, EX1	54.48	219	35.57	143	5.39*	0.000	[-0.257, -0.122]
	215, EX1	60.93	131	37.67	81	4.82*	0.000	[-0.324, -0.141]
	187, EX1	47.59	89	35.83	67	2.31*	0.021	[-0.217, -0.018]
	187, EX2	48.13	90	32.09	60	3.17*	0.002	[-0.258, -0.062]
C4.5 [†]	402, EX1 + EX2	49.75	200	31.84	128	5.17*	0.000	[-0.246, -0.112]
	402, EX1	56.22	226	35.57	143	5.87*	0.000	[-0.274, -0.139]
	215, EX1	60.93	131	37.67	81	4.82*	0.000	[-0.324, -0.141]
	187, EX1	51.34	96	35.83	67	3.02*	0.002	[-0.254, -0.056]
	187, EX2	43.32	81	32.09	60	2.24*	0.025	[-0.21, -0.015]
Mean		52.94		34.60				

*The proportions' difference is significant at the $P < 0.05$ level.

[†]Refers to the 'Data mining-generated classifier' column.

Table 2 A comparison of the accuracy of the classifiers generated by the data mining algorithms with the accuracy of the Maximum probability classifier, using a proportions test

		Data mining-generated classifier		Maximum probability classifier		z statistic	P-value	Confidence interval (95%)
		Accuracy (%)	n	Accuracy (%)	n			
Naïve Bayes [†]	402, EX1 + EX2	56.72	228	40.80	164	4.52*	0.000	[-0.227, -0.091]
	402, EX1	54.48	219	50.50	203	1.13	0.258	[-0.109, 0.029]
	215, EX1	60.93	131	53.95	116	1.46	0.143	[-0.163, 0.023]
	187, EX1	47.59	89	46.52	87	0.21	0.836	[-0.112, 0.09]
	187, EX2	48.13	90	35.83	67	2.41*	0.016	[-0.222, -0.024]
C4.5 [†]	402, EX1 + EX2	49.75	200	40.80	164	2.55*	0.011	[-0.158, -0.021]
	402, EX1	56.22	226	50.50	203	1.63	0.104	[-0.126, 0.012]
	215, EX1	60.93	131	53.95	116	1.46	0.143	[-0.163, 0.023]
	187, EX1	51.34	96	46.52	87	0.93	0.352	[-0.149, 0.053]
	187, EX2	43.32	81	35.83	67	1.48	0.139	[-0.174, 0.024]
Mean		52.94		45.52				

*The proportions' difference is significant at the $P < 0.05$ level.

[†]Refers to the 'Data mining-generated classifier' column.

Table 3 A comparison of the data mining-generated classifiers with the Random and Maximum probability classifier, using the BCR* and HMSS* measures

		Data mining-generated classifier		Random Classifier		Maximum probability classifier	
		BCR*	HMSS*	BCR*	HMSS*	BCR*	HMSS*
Naïve Bayes [†]	402, EX1 + EX2	0.643	0.503	0.500	0.341	0.5	0
	402, EX1	0.613	0.467	0.500	0.322	0.5	0
	215, EX1	0.648	0.539	0.500	0.312	0.5	0
	187, EX1	0.550	0.325	0.500	0.321	0.5	0
	187, EX2	0.599	0.420	0.500	0.338	0.5	0
C4.5 [†]	402, EX1 + EX2	0.595	0.450	0.500	0.341	0.5	0
	402, EX1	0.623	0.502	0.500	0.322	0.5	0
	215, EX1	0.607	0.401	0.500	0.312	0.5	0
	187, EX1	0.569	0.372	0.500	0.321	0.5	0
	187, EX2	0.557	0.410	0.500	0.338	0.5	0
Mean		0.600	0.439	0.500	0.327	0.5	0

[†]Refers to the 'Data mining-generated classifier' column.

BCR*, multiple-value Balanced Correctness Rate; HMSS*, multiple-value Harmonic Mean of Sensitivity and Specificity.

probability classifier neither raise nor reduce Entropy ($H_{\text{classifier}} = RCI = 0$). This occurs because their outcomes are no different from the original distribution of severity grades, as determined by the physicians.

Comparison of the results of the DM versus the agreement between the physicians

We compared the results of the predictions made by the DM-generated classifiers with the level of agreement among the two physicians, which can be considered as asking one physician to 'predict' the classification made by the other. Thus, this level of agreement can be considered as a type of 'accuracy measure' for the ability of each physician to 'predict' the other's classification. We applied proportions tests to compare the accuracy produced by the DM-generated classifiers and by the physicians' agreement.

This test was applied to the set of the last 187 case records, because only these records were classified for severity grade by both physicians.

The rate of complete agreement (i.e. both assign exactly the same severity grade) between the two physicians was 39.5%. The accuracy levels of the classifiers generated by the Naïve Bayes DM algorithm were 47.5% when learning from 187, EX1 and 48.1% when learning from 187, EX2; the accuracy levels of the classifiers generated by the C4.5 algorithm were 51.3% and 43.3%, respectively. Furthermore, we used a quadratic weighted kappa statistic to measure their inter-rater agreement. We used a weighted kappa measure, because we wanted to penalize severity score disagreements in terms of the scores' absolute difference, as opposed to the use of an unweighted kappa statistic, which would have treated all score differences equally (thus considering a severity grade of 4 and a severity grade of 2 to be equally distant from a severity grade

Table 4 Results of the comparison between prior and posterior Entropy using the $H_{\text{classifier}}$ and RCI measures

		H_d (prior Entropy)	H_o (posterior Entropy)	$H_{\text{classifier}}$ ($H_d - H_o$)	RCI ($H_{\text{classifier}}/H_d$)
Naïve Bayes	402, EX1 + EX2	1.7500	1.5236	0.2264	0.1294
	402, EX1	1.6822	1.5044	0.1778	0.1057
	215, EX1	1.6461	1.4256	0.2206	0.1340
	187, EX1	1.6499	1.4778	0.1721	0.1043
	187, EX2	1.7044	1.5505	0.1540	0.0903
C4.5	402, EX1 + EX2	1.7500	1.5994	0.1506	0.0860
	402, EX1	1.6822	1.5179	0.1643	0.0977
	215, EX1	1.6461	1.3865	0.2597	0.1577
	187, EX1	1.6499	1.5498	0.1002	0.0607
	187, EX2	1.7044	1.6205	0.0839	0.0492
Mean		1.6865	1.5156	0.1709	0.1014

RCI, Relative Classifier Information.

of 1). The quadratic penalty, which is a common weighting method, seemed to best reflect also the clinical meaning of the different scores. The quadratic weighted kappa for the two physicians on the 187 patients records which were both examined was positive and statistically significant ($\kappa = 0.381$, $P < 0.05$).

In addition, we took advantage of the fact that the severity grades are ranked on an ordinal scale to count each of the results deviating from the severity grade determined by a physician by only one or less as a hit. We indicate this approximation by the label 'Approximate (± 1) grade'. The rate of 'Approximate (± 1) grade' agreement between the two physicians was 88.2%. The 'Approximate (± 1) grade' accuracy levels of the classifier generated by the Naïve Bayes DM algorithm were 86.1% when learning from 187, EX1 and 88.2% when learning from 187, EX2; the 'Approximate (± 1) grade' accuracy levels of the classifier generated by the C4.5 algorithm were 85.5% and 86.6%, respectively.

These results indicate a significant improvement in the accuracy, when using the 'Approximate (± 1) grade' label, as compared with the level of accuracy when using the complete agreement ('Actual grade'). A difference exists between (1) the physicians' agreement rate (i.e. 'accuracy'), when we only accepted complete agreement as a hit (Actual grade) and (2) the physicians' agreement rate when we used an Approximate (± 1) grade. However, the accuracy of the DM-generated classifiers remains similar to the physicians' agreement rate. Thus, the classifiers do indeed embody learning and succeed in mimicking the physicians' severity assignments for the same patient records, including the physicians' understanding of the ordinal nature of the severity-grade scale.

Using an approximate accuracy measure

We also applied the approximate-grade measure to the full data set and used a proportions test to detect significant differences in all of the classified case categories.

There was a significant difference between the accuracy outcomes when using the 'Approximate (± 1) grade' label and those of the Actual grade, with a higher accuracy when using the 'Approximate (± 1) grade'.

Overall, the average accuracy (for all of the results, using both DM methods), when accepting only exact agreement, is 52.94%, while using the 'Approximate (± 1) grade' measure yields an average accuracy of 85.42%.

Aggregation of severity grades

Within the framework of our research, we chose to aggregate the severity grades in such a way that reduces their number. We decided to aggregate them into either two or three distinct grades (or classes of grades). After studying all of the possible combinations for such an aggregation, each one of these possible combinations was compared with all the rest of the combinations, including the base case of four grades, according to accuracy and the values derived by the HMSS* and BCR* measures.

We will present in details only one example of such a comparison – between two grades of severity (1, 2 + 3 + 4) (i.e. the aggregation of grades 2, 3 & 4 into one grade, while using grade 1 as is) and the full four-grade severity scale (Table 5).

Table 5 shows that there is a significant difference between the accuracies achieved using the two severity scales described above, with a higher accuracy when using the (1, 2 + 3 + 4) scale than when using the full four-grade one.

We also applied more sophisticated measures (HMSS*, BCR*). Using these measures further emphasized the tendency demonstrated in the results of the proportions test. In both the HMSS* and BCR* measures, the outcomes when using two grades of severity (1, 2 + 3 + 4) are higher than when using the four-grade severity scale, with the exception of one case only.

In general, we can say that the results of comparing all of the combinations for the aggregation of the severity grades indicate a preference for a smaller number of grades. That is to say that it is preferable to use two grades, rather than three or four grades, and to use three grades, rather than four. But these results are not conclusive and depend primarily on the method used in each case to combine the four grades into a smaller number of grades.

It is worth mentioning that the precise manner of the aggregation of severity grades into more abstract classes significantly affects the resulting values of different classification measures.

Table 5 A comparison of the accuracy of the classifiers when using a two-grade scale (1, 2 + 3 + 4) with their accuracy when using the full four-grade scale, using a proportions test

		2 grades (1, 2 + 3 + 4)		4 grades		z statistic	P-value	Confidence interval (95%)
		Accuracy (%)	n	Accuracy (%)	n			
Naïve Bayes	402, EX1 + EX2	72.64	292	56.72	228	4.72*	0.000	[-0.224, -0.094]
	402, EX1	71.14	286	54.48	219	4.89*	0.000	[-0.232, -0.101]
	215, EX1	74.42	160	60.93	131	2.99*	0.003	[-0.222, -0.047]
	187, EX1	67.91	127	47.59	89	3.98*	0.000	[-0.301, -0.105]
	187, EX2	69.52	130	48.13	90	4.2*	0.000	[-0.311, -0.117]
C4.5	402, EX1 + EX2	70.65	284	49.75	200	6.05*	0.000	[-0.275, -0.143]
	402, EX1	70.65	284	56.22	226	4.25*	0.000	[-0.21, -0.078]
	215, EX1	73.95	159	60.93	131	2.88*	0.004	[-0.218, -0.043]
	187, EX1	71.66	134	51.34	96	4.04*	0.000	[-0.3, -0.107]
	187, EX2	71.12	133	43.32	81	5.43*	0.000	[-0.374, -0.182]
Mean		71.37		52.94				

*The proportions' difference is significant at the $P < 0.05$ level.

This was especially clear in those cases in which an unbalanced division of severity grades is produced, as when using two grades of severity with the combination (1 + 2 + 3, 4). In such cases, it is important to use not only the Accuracy measure (which does provide high results), but also the HMSS* and BCR* measures. The use of these more sensitive measures results in such cases in a very low score, because of the fact that obtaining a higher degree of accuracy for the severity grades characterizing the majority population is at the expense of obtaining a much lower accuracy in the case of the severity grade class that characterizes the minority population.

Comparison of the performance of the C4.5 and Naïve Bayes algorithms

As stated above, in this research we used two DM algorithms C4.5 and Naïve Bayes. When comparing the results obtained from both algorithms (e.g. as can be seen in Table 1, and as was further validated by testing for significant differences between the outcomes of both algorithms), it was found that, in general, there was no significant difference between both DM algorithms, despite their being conceptually different and unrelated to each other.

There was only one case in which Naïve Bayes was significantly preferable to C4.5 (using the advanced measures, although the standard measures results in high scores for both algorithms) and that was when creating two grades by using the combination (1 + 2 + 3, 4). Such a combination results in a very unbalanced division (c. 95% in one grade and c. 5% in the other).

Learning from the Consensus set

As this research has involved two physicians whose classifications were defined as the 'gold standard', we tested whether there was a significant difference in the accuracies obtained when learning only from those cases having complete agreement, in which there was a consensus between both physicians, versus learning from all of the recorded cases classified by both physicians (even when there was no consensus), each time classified by only one of the physicians. The Consensus set, in which both physicians agreed on every case, includes 74 cases out of the 187 classified by both of them.

There was a significant difference between the accuracy obtained when learning from the Consensus group and when learning from the EX2, 187 group, both in a four-grade scale and in a two-grade severity ranking; This difference held true when using either of our DM algorithms, with the accuracy being the highest when learning from the Consensus group. This difference was evident also when learning from the EX1, 187 group, although it was not always significant.

Discussion

Analysis of the results

This study has shown that the results obtained by using the DM-generated classifiers, training them by learning from the physicians' classification of the patient records, are significantly superior, with respect to all measures, to the results achieved using algorithms that are based only on the severity-grade distribution. Moreover, we have demonstrated this conclusion to be independent of the DM algorithms used and of the data (sub)sets used in each experiment.

Because of the imbalance found in the distribution of the severity grades, we used, in addition to the accuracy measure, also the HMSS* and BCR* measures. These more delicate measures demonstrate that the results produced by the DM-generated classifiers are superior to the results of an algorithm that exploits the lack of balance between the various severity grades, by assigning all cases to the severity grade that has the highest prevalence in the patient population, even when the accuracy of that algorithm approaches the accuracy of the DM-generated classifiers. The use of integrated measures based on both sensitivity and specificity, in addition to the accuracy measure, seems indicated in most cases in which the target classes in the data set is unbalanced, as is common in medicine. The reason is that using a maximum probability classifier does not permit predicting any minority class, with dire consequences, especially in the medical domain, in which detection of such classes is often of prime importance. It also precludes any meaningful case-based *learning*, by definition, as the classifier's behaviour does not depend on any features of the patients, but rather on the prior distribution of the severity grades.

Moreover, we showed that the DM-generated classifiers reduce the prior Entropy, and thus reduce the level of uncertainty regarding the grade of severity. This is an indication that the results of the classifier really embody additional information, although the classifier need not necessarily be accurate regarding the category being predicted. This observation enhances the value of a DM-generated classifier for use as a basis for a triage decision support system. Such a system can exploit the distribution of severity grades, given each severity grade predicted by the classifier, to optimize its recommendation (typically, by also assuming some cost function for assigning the patient into severity grade i , when the real severity grade is j). Thus, the decision to classify a certain case as having a certain severity grade, might be based both on the distribution of severity grades implied given the output of the classifier, and on the *utility* (or cost) function, defined by the physicians, of misclassifying certain severity grades; the misclassification costs need not be uniform, nor symmetric.

Comparison of the DM results and the agreement between the physicians

The DM-generated classifiers predicted a physicians' classification as least as accurately as the agreement with the other physician. The relatively low level of agreement amongst the physicians might suggest an upper-bound 'glass ceiling' to the possible achievable level of accuracy when comparing the system to a group of physicians. Thus, it might be advisable to compare the DM-generated classifiers only to the severity grade assigned by a Majority of the physicians, as was done in previous evaluations of medical expert systems [23].

By exploiting the ordinal nature of the severity grade scale, and considering as an 'agreement' also a disagreement of up to one grade difference, we improved agreement between physicians from 39.5% to 88.2%. The respective improvement, when permitting such a relaxation, in the results of the DM-generated classifiers when compared to either physician, was from a mean of 48% to a mean of 87% accuracy. This analogous behaviour further proves that the classifiers do indeed embody learning and succeed in mimicking the physicians' severity assignments, including an understanding of the ordinal nature of the severity-grade scale.

Accuracy when allowing a one-grade discrepancy in the classification

As we saw, even a moderately performing classifier can be quite accurate when allowing deviations of up to one grade in the level of severity. Thus, when a classifier predicts a low severity grade (e.g. grade one), the likelihood that the patient's severity grade is high (e.g. grade three or more) is very low, and vice versa. This finding fits well with similar outcomes from the research of Manos *et al.* [24], which discussed severity grades assigned by a medical team to patient cases during triage. According that study, accuracy when allowing for a deviation of one severity grade or less was significantly greater than the accuracy using the actual grade. We conclude that the four-grade severity scale might include an internal overlap amongst adjacent severity grades.

Determining the optimal number of severity grades

The manner in which, in our experiments, we aggregated the original four severity grades into a smaller number of categories was very significant, with respect to the performance of the automated classifiers. This was especially evident when the aggregation led to highly unbalanced data sets, requiring the use of the HMSS* and BCR* measures, and not only the accuracy measure (which produces illusively high scores), to test both the sensitivity and the specificity.

The reason for the classification-performance enhancement for a smaller number of categories (besides the possibility that classification into a smaller number of classes might be easier) seems to be due to the aforementioned inherent blurring among the severity grades (especially in the middle grades). Thus, in several of the cases, aggregating the grades into a smaller number of categories has successfully reduced some of this internal blurring by uniting the two least distinct categories. However, the improvement, if any, depends on the manner in which the grades are combined, that is, which grades are combined into a single category.

The inherent ambiguity of severity grades

Our class-aggregation experiment demonstrated a preference (from the point of view of the automated classifiers) for a small number of categories, suggesting that there is some blurring among the severity grades, especially among the second grade and its neighbouring first and third grades. This insight is strengthened by the results of the classifiers and of the physicians when using a relaxed accuracy measure, which allows a deviation of up to one severity grade, thus significantly improving the accuracy, both of the DM-generated classifiers and of the physicians' Consensus (agreement with each other). Both experiments suggest an ambiguity in the severity grades assigned by the physicians, as both the use of the relaxed measure as well as the aggregation of severity grades into a smaller number of categories cause the middle grades to be aggregated into one of the adjacent grades.

Our results fit well with similar outcomes in the study of Manos *et al.* [24]. In that study, the authors also found that the agreement rate between the medical personnel was lower in the middle grades.

In addition to an ambiguity in the severity grades definition, it is also possible to explain the 'middle-grade ambiguity' phenomenon by the fact that it is easier for the physicians to determine whether a patient is in a mild or a serious state than to distinguish between the medium grade (second grade, in our case) and either one of its neighbours on the severity scale.

Comparison of the Naïve Bayes and C4.5 algorithms

For the purposes of this study, we used two DM algorithms (Naïve Bayes, C4.5). By comparing the outcomes produced by both these classifiers, it was found that, in general, there is no significant difference in their performance, despite their methodological differences. Since these algorithms are essentially independent of one another, we recommend the use of both of them in tandem.

There was one exceptional case in which the Naïve Bayes algorithm is significantly preferable to the C4.5 algorithm, and that is when the data were divided into two unbalanced categories (in our study, 95% in one set and 5% in the other). In such types of cases, it might be preferable to use the Naïve Bayes algorithm. It is important to note this result, because such cases are often found in the field of medicine, such as when there is a small percentage of patients compared with the percentage of healthy people.

Using the Consensus set

We found, during the course of our research, that it is preferable to learn from predetermined, Consensus-based case categories (and to apply the data learned to other cases within the same category), rather than learning from a set of all of the cases, classified by physicians. In the light of these findings, we conclude that it would be preferable to create classifiers that rank the patients by severity grade on the basis of learning from the physicians' Consensus, rather than from one physician's classification.

Advantages and limitations of this study

The main advantage of our method (learning by data mining) is that it is automatic and does not require manual engineering to build rules and flow charts, which requires time and resources to create and maintain knowledge.

Another advantage, implied by our data mining methodology, is that the classifiers continuously learn from accumulating cases, and thus continually improve over time, as the database (the number of examples/cases) expands.

As far as we know, no similar research has been done thus far on the classification of degrees of severity by means of a machine learning methodology, based on physicians' rankings. Most of the existing methods for performing triage are based on manual algorithms produced by a physician's consensus, etc., determining the appropriate degree of severity of each case, as in the Canadian method [7] and the method developed by the Manchester Triage Group [13].

One limitation of this study might be that although it includes a greater number of cases than found in prior research [24] (in regard to comparisons of the physicians' rankings), it is still a relatively small number of cases (402) upon which to base a machine learning approach. Perhaps the study of a larger number of cases might yield better results.

Another limitation of this study is that we limited ourselves to the severity rankings of only two physicians, bearing in mind the fact that we found the outcomes when learning from the Consensus to be better than when learning from one physician. We recommend the use of a greater number of physicians in future research and focusing on the Consensus classification of those physicians. In particular, this approach will enable comparing the classifiers predictions to that of the majority of physicians – a much more robust measure.

Conclusions

This paper has presented encouraging results, indicating that it is possible to automatically learn the physicians' method of ranking patients' cases by the degree of severity, on the basis of existing

data at the triage stage, and thus develop a classifier for severity ranking, using DM algorithms. We recommend that further studies be performed using this new methodology, and that other classifiers be developed to rank severity grades during triage, by means of machine learning methods based on physicians' assignments of severity grades to known cases.

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